Please show all work for full credit

1. Find the values of m for which $\phi(x) = e^{mx}$ is a solution to the differential equation

$$\frac{d^2y}{dx^2} + 6\frac{dy}{dx} + 5y = 0$$

2. Solve $\frac{dy}{dx} = x^2 e^{-4x} - 4y$.

3. Classify the differential equation $x\left(\frac{d^2y}{dx^2}\right)^3 + \frac{dy}{dx} + xy = 0$ (2 pts each).

a. linear or nonlinear (circle one)

b. order 1 , order 2. or order 3 (circle one)

c. ODE or PDE (circle one)

d. the independent variable is ____.

e. the dependent variable is ____.

4. Does the Theorem of Existence and Uniqueness guarantee that given initial value problems have unique solutions? (2.5 pts each)

	Equation	**Condition**	**Yes or No**
a.	$\frac{dy}{dx} = y^4 - x^4$	$y(0) = 7$	
b.	$\frac{dy}{dt} - ty = \sin^2 t$	$y(\pi) = 5$	
c.	$\frac{dx}{dt} + \cos x = \sin t$	$x(\pi) = 0$	
d.	$\frac{dy}{dx} = \sqrt{y-1}$	$y(2) = 1$	

5. Classify the differential equation $\theta\, dr + (3r - \theta - 1)\, d\theta = 0$. Select all that apply.

a. Linear if r is the dependent variable.

b. Linear if θ is the dependent variable.

c. Separable

d. Exact

e. Bernoulli if r is the dependent variable.

f. Bernoulli if θ is the dependent variable.

g. Homogeneous

6. Solve $y' = \frac{y}{y+1}$, $y(1) = 1$

7. Solve $\left(ye^{xy} - \frac{1}{y}\right) dx + \left(xe^{xy} + \frac{x}{y^2}\right) dy = 0$, $y(1) = 1$

8. Find an integrating factor to make the equation exact and solve

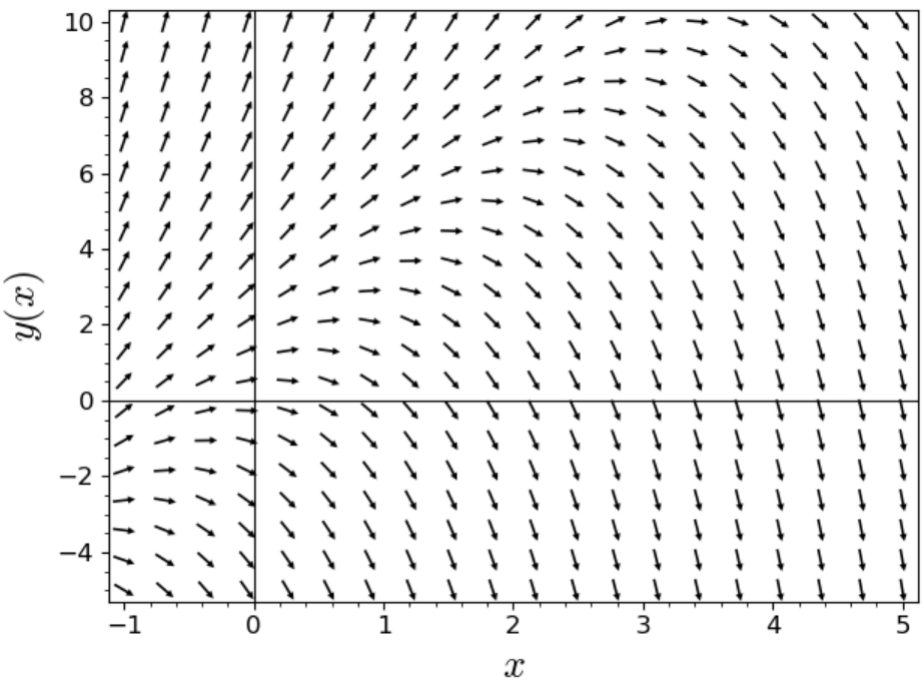
$$\left(2y^2 + 2y + 4x^2\right) dx + (2xy + x)dy = 0$$

9. Solve $\frac{dy}{dx} + y = e^x y^{-2}$. Your answer should be in explicit form.

10. Solve $\frac{dy}{dx} = \frac{y(\ln y - \ln x + 1)}{x}$. Your answer should be in explicit form.

11. Use the Euler Method to find $y(0.1)$ and y(0.2) for the DE $y' = 2x - 3y$ using $h = 0.1$ and $y(0) = 1$.

12. Find the DE corresponding to the given direction field.



a. $y' = 2x$

b. $y' = 2 + 3x$

c. $y' = y - 3x$

d. $y' = \sin x \cos y$

e. $y' = \cos x \cos y$